Placing Defects Where You Ask: Generating Defects at Chosen Sites in a Model Glass with a Diffusion Model

Midterm report

Stacy Cheruto Nicholaus Ladislaus Abubakar Chilala
Gentille Uwera Michel Sangwa Boniface Godwin[†]
College of Engineering, Carnegie Mellon University Africa
{ccheruto, eladisla, achilala, guwera, sbmichel, bgodwin}@andrew.cmu.edu
[†]Teaching assistant supervising this project

Abstract

Glasses contain *defects*, small groups of atoms arranged differently from the rest, which decide how a glass breaks, ages or absorbs light. Studying one needs many realistic copies, but today one waits for it in slow melt-and-cool simulations or inserts it by hand, which can leave unnatural surroundings. We train a diffusion model, a neural network that turns random atom positions into realistic structure, to rebuild a small region of an existing glass with a requested defect at its centre. On a two-dimensional model glass, over three training runs, it creates atoms with too few neighbours 3.5 times as often as hand insertion (0.22 vs. 0.06) and clusters of small atoms 1.5–2.4 times as often (0.45–0.71 vs. 0.30), with surroundings whose atom energies are closer to natural ones.

1 Introduction

Motivation. A defect's behaviour depends on the atoms around it, so studies of how glasses deform, age or absorb light need many realistic examples of the same defect at chosen places. Melt-and-cool (melt-quench) molecular dynamics (MD) waits for the defect to form, which is slow; hand insertion is fast but can leave unnatural surroundings. Scientists who build defect libraries, test theories of glass failure or train machine-learned force fields would benefit from a tool that places a requested defect at a requested site. Diffusion models such as DM2 [15] generate realistic glasses, but are controlled by whole-sample properties such as the cooling rate.

Objectives. Like image inpainting [10], our model fills in a region of an existing glass (the rest held fixed) with a requested defect at its centre (Task B). Objectives:

O1 **Placement:** after relaxation, the requested defect is at the site in more of 256 requests than with hand insertion or local melt-quench.

O2 **Realism:** atom energies around a placed defect are closer to natural ones than with hand insertion (local W1), approaching the natural floor.

O3 **Reliability:** O1 and O2 hold over three training runs and under changes to the evaluation settings.

We start with a two-dimensional model glass, where every step is cheap to check, before moving to 3D silica.

2 Related work and background

Literature review. Diffusion models [7, 12] now generate crystals [8, 14, 16] and glasses. Our key paper is **DM2** [15]: a network learns to predict random shifts added to atom positions, and glasses

are built by repeating this clean-up with small random kicks. We keep its training target and sampler and add what it lacks, a defect label and held atoms. For local control, RePaint [10] fills a masked image region and classifier-free guidance [6] makes a model follow a label more strongly; materials work fills in crystals around a fixed host [17].

Background: metric and baselines. DM2 judges realism by comparing distributions of structural quantities in generated and MD-made glasses; our local W1 applies this idea to the region around one defect. DM2's unlabelled model is our "chance" baseline and RePaint our baseline for building a whole new box (Task A). Hand insertion and melt-quench MD are how defects are made in simulations today, so they are our main baselines.

3 Methodology

3.1 Dataset

We generate our data with LAMMPS [13]: the two-dimensional Kob–Andersen glass [4, 9], with large A (65%) and small B (35%) atoms interacting through Lennard-Jones forces in a periodic box; σ, the size of an A atom, is the unit of length. Preparation: melt at $T = 2.0$, cool by 10^{-2} per time unit to $T = 0.01$, minimise, reject crystallised or faulty glasses, and label defects with rules fixed on training glasses. The main set has 800/100/100 training/validation/test glasses of $N = 256$ atoms (box side 14.6σ); 1600/200/200 glasses with $N = 64$ serve a symmetry test. With neighbours counted within 1.4 contact distances, D1$^-$ is an A atom with at most 5 neighbours (1.7 per glass), D1$^+$ one with at least 8 (1.4 per glass), and D2 a B atom with at least two B neighbours within 1.2σ (10 per glass). A library of 10,545 natural defects from the training glasses supplies hand insertion.

3.2 Model description

We describe the layers of the denoising network, the guidance used when sampling and the architectures compared (Figure 1, Table 1).

Input layer. The input is one glass at noise level σ: for each of the N atoms a position $x_i \in \mathbb{R}^2$ in a periodic box, a type $t_i \in \{\mathrm{A},\mathrm{B}\}$, a flag $m_i = 1$ if the atom is held fixed, and a label token c_i, "unlabelled" except at the requested site, which carries the requested defect (or "null" when dropped). With σ encoded as 64 random Fourier features of $\log\sigma$, the starting feature vector of atom i is

$$h_i^0 = \mathrm{MLP}\big(E_t(t_i) + E_m(m_i) + E_c(c_i) + \mathrm{MLP}(\mathrm{Fourier}(\log\sigma))\big) \in \mathbb{R}^{128}, \tag{1}$$

where E_t, E_m, E_c are learned look-up tables. Positions enter only through the graph.

Graph layer. Atoms closer than $r_c = 2.5\sigma$ are linked (about 24 neighbours), with r_{ij} the shortest vector from atom i to j across the periodic box. Distances are expanded in 16 Gaussian radial basis functions and multiplied by a smooth cut-off $f_c(r) = \frac{1}{2}[\cos(\pi r/r_c) + 1]$.

EGNN layers. Four layers (Figure 1b) pass messages between linked atoms, update the features and produce one vector per atom:

$$\begin{aligned} m_{ij} &= \phi_e\big(h_i, h_j, \mathrm{rbf}(|r_{ij}|)\big)\, f_c(|r_{ij}|), \qquad h_i \leftarrow h_i + \phi_h\Big(\mathrm{LN}(h_i),\ \tfrac{1}{K_0}\textstyle\sum_j g(m_{ij})\, m_{ij}\Big), \\ v_i^\ell &= \tfrac{1}{K_0}\textstyle\sum_j r_{ij}\, \phi_x(m_{ij})\, f_c(|r_{ij}|). \end{aligned} \tag{2}$$

ϕ_e ($272 \to 128 \to 128$), ϕ_h ($256 \to 128 \to 128$) and ϕ_x ($128 \to 128 \to 1$) are two-layer SiLU networks, g a sigmoid gate, LN layer normalisation and $K_0 = 24$. Messages depend only on distances; v_i^ℓ, a weighted sum of the vectors r_{ij}, turns with the glass [11]. Each layer has 117,890 parameters.

Output layer. The output $d_i = (1 - m_i)\sum_\ell v_i^\ell \in \mathbb{R}^2$ estimates how far the noise moved atom i (no activation; zero for held atoms).

Classifier-free guidance. When sampling, the network runs with and without the site label (Figure 1c) and combines the outputs, $\tilde{d} = (1 + w)\, d_\theta(\cdot, c) - w\, d_\theta(\cdot, \varnothing)$; w sets how strongly it follows the label.

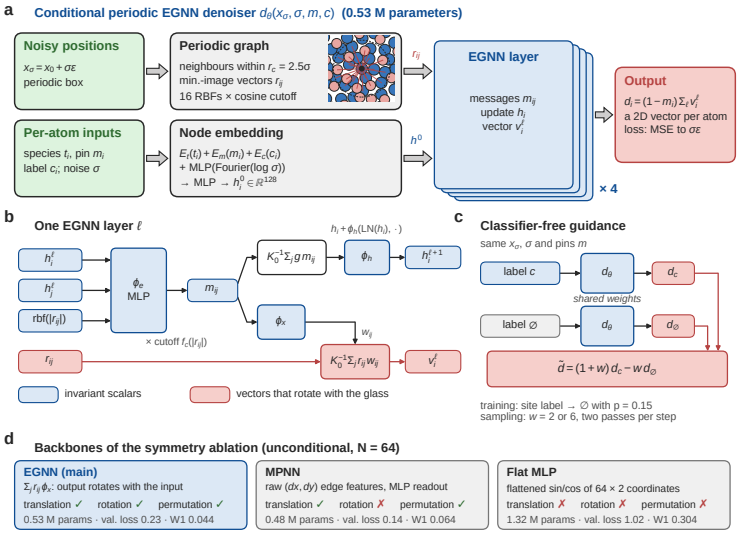


Figure 1: The networks. **(a)** The conditional EGNN denoiser: noisy positions define a periodic graph, per-atom inputs define starting features, and four EGNN layers turn both into one arrow per atom. **(b)** One EGNN layer: blue items do not change when the glass is rotated; red items rotate with it. **(c)** Classifier-free guidance runs the same network with and without the site label. **(d)** The three networks of the symmetry test.

Table 1: Deep learning models; $N \times \{\text{layer}\}$ is a layer repeated N times. Values in brackets were tried. Fixed: 128 features, cutoff 2.5σ, 16 radial basis functions, AdamW, learning rate 2×10^{-4}, batch size 16.

Model	Hidden layers	Parameters / hyper-parameters	Size
EGNN_cond (main) and EGNN_uncond	$1 \times \{$input embedding$\}$, $4 \times \{$EGNN layer$\}$, output sum	labels (yes, no); training noise $\sigma_{\max}$ (0.5, 1.0); region R_{out} (4σ, 3–6σ); seed (0, 1, 2); guidance w (0–8); sampling steps (100, 300, 900, 2900); N (64, 256)	0.53 M
MPNN	$1 \times \{$input embedding$\}$, $4 \times \{$MPNN layer$\}$, {Dense 128, Dense 2}	$N = 64$; no label	0.48 M
Flat MLP	$4 \times \{$Dense 512$\}$, Dense 128	$N = 64$; no label	1.32 M

Model architectures (Table 1): EGNN_cond is the main model, EGNN_uncond the same network without labels; to test built-in symmetry, an MPNN [5] that also sees raw (x, y) neighbour vectors and a flat MLP were trained on 64-atom glasses (Figure 1d).

3.3 Loss function and training

As in DM2, noise $x_\sigma = x_0 + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, $\sigma \sim \mathcal{U}(10^{-3}, \sigma_{\max})$ is added to the free atoms U and the network predicts the shift, with a mean squared error loss:

$$\mathcal{L}(\theta) = \mathbb{E}\left[\frac{1}{|U|}\sum_{i \in U} \left\| d_\theta(x_\sigma, \sigma, m, c)_i - \sigma\varepsilon_i \right\|^2\right]. \tag{3}$$

Examples are whole boxes, patches or filled-in regions (50/30/20%), centred on a natural defect half of the time, with the label set to "null" 15% of the time for guidance. We use the 0.80:0.10:0.10 split (validation for over-fitting checks and settings, test only for results), PyTorch, AdamW and 50,000 updates (about 1,000 passes; 4 hours on a T4 GPU).

3.4 Evaluation metrics

Success answers "did I get the defect I asked for, where I asked?": after relaxation with FIRE [2] (all atoms free), the atom of the requested type nearest the site lies within 0.5σ and carries the requested label. **Local W1** answers "do the surroundings look natural?": the Wasserstein-1 distance between the energies of atoms within 3σ of the site and those around natural defects of that type. Natural

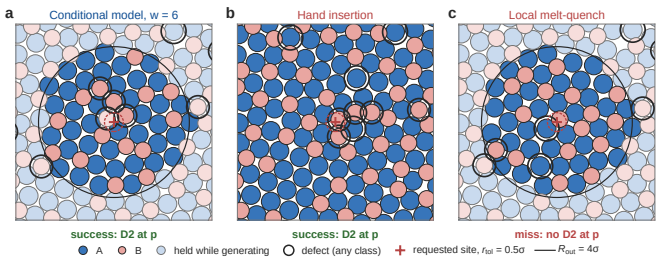


Figure 2: One Task B request answered by each method in our pipeline, after relaxation (pale atoms held fixed; rings mark defects): **(a)** the model and **(b)** hand insertion place a D2 defect at the site; **(c)** local melt-quench does not.

training versus test defects give the *natural floor* (D2 0.013, D1$^-$ 0.016). Intervals are 95% bootstrap CIs over 256 requests.

3.5 Experimental iterations

We trained 13 models (Table 1): two unlabelled, three seeds for each of two noise ranges, three for the symmetry test and two with a range of region sizes. DM2's noise range ($\sigma_{\max} = 0.5$) left the network unprepared for the $\sigma = 1$ start of sampling; retraining with $\sigma_{\max} = 1.0$ raised D2 success from 0.54 to 0.79 (seed 0, $w = 6$). We cut the sampler from 2,900 to 100 steps (longer runs over-cool the glass). The EGNN made more natural glasses than the MPNN (W1 0.044 vs. 0.064); the flat MLP did not learn.

4 Baselines and extensions

Baselines (implemented and run by us). *Hand insertion*, today's common practice: paste a natural defect with its neighbours within 1.5σ at the site, then relax. *Local melt-quench*, the MD route: freeze atoms beyond R_{out} and melt and cool only the region in LAMMPS. *Chance*: fill the region with the unlabelled model. Task A adds clamping and RePaint [10]. All methods answer the same 256 requests.

Baseline reproduction evidence. All baseline numbers in Table 2 come from our own runs; Figure 2 shows one request answered by each method. Re-running nine rows with the same seed reproduced eight exactly, and re-running DM2's published silica model confirmed its local structure (99.3% of Si atoms four-fold).

Extensions and experiments. Beyond DM2 we add defect labels, held atoms, a noise-level input and guidance, and test: (i) the baselines; (ii) guidance chosen per model on validation glasses; (iii) the rebuilt region's size; (iv) physical checks (structure, survival in MD); (v) larger glasses and other cooling rates; (vi) 3D silica.

5 Results

Main comparison (Table 2). The model beats both baselines for D2 and D1$^-$ (O1), with local W1 1.2–1.5 times the floor for D1$^-$ and 0.045–0.056 for D2 at $w = 2$ (hand insertion 0.087; O2). No method places D1$^+$: a natural D1$^+$ transplanted with its 47 nearest atoms vanishes 97% of the time. The ranking held when we changed the defect rules, success radius or realism window (O3).

Choosing w for each model. The D2 runs follow the label with different strength despite near-equal validation losses. Picking on validation glasses the smallest w with success ≥ 0.70 ($w = 4, 8, 8$) gave test success 0.74 ± 0.04, half the spread of fixed $w = 6$ (0.71 ± 0.08; Figure 3a).

Region size. A larger region lowers W1 (0.026 at $R_{\text{out}} = 5\sigma$, 0.056 at 4σ). Shell by shell (Figure 3b), the mismatch stays about 3 times the floor up to the region's edge (hand insertion:

Table 2: Task B: success after relaxation and local W1 (lower is closer to natural). Model rows: mean ± standard deviation over three training runs; baselines: estimate with 95% CI. D2 uses $\sigma_{\max} = 1.0$; D1$^-$ and D1$^+$ use $\sigma_{\max} = 0.5$. †2,900-step sampler. ‡Seed 0 only. $^\S w$ chosen for each model on validation glasses.

	D2 (B-rich)		D1$^-$ (too few neighbours)		D1$^+$
Method	success	local W1	success	local W1	success
Chance (no label)†	0.03	0.062	0.02	0.053	0.00
Local melt-quench	0.05 [0.03, 0.08]	0.065	0.00 [0.00, 0.00]	0.024	0.01
Hand insertion	0.30 [0.24, 0.36]	0.087	0.06 [0.04, 0.09]	0.032	0.02
Model, $w = 2$	0.45 ± 0.11	**0.051 ± 0.006**	**0.22 ± 0.02**	**0.022 ± 0.002**	0.03‡
Model, $w = 6$	0.71 ± 0.08	0.072 ± 0.016	0.21 ± 0.04	0.038 ± 0.004	0.04‡
Model, w per model§	**0.74 ± 0.04**	0.076 ± 0.013	—	—	—
Natural floor	—	0.013	—	0.016	—

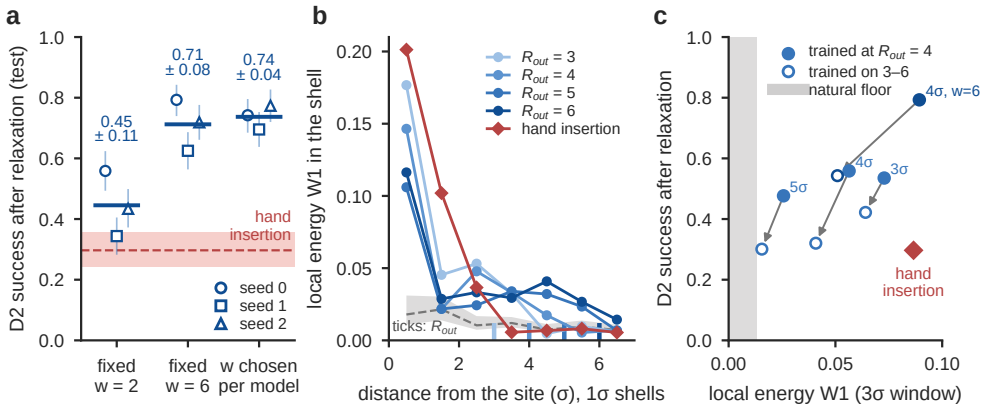


Figure 3: Guidance and region size (Task B, D2). **(a)** Test success per training run at fixed w and at w chosen per model on validation glasses; bars: means; red: hand insertion. **(b)** Local energy W1 in 1σ shells around the site for $R_{\text{out}} = 3$–6σ (axis ticks), hand insertion and the natural floor (grey). **(c)** Model trained at $R_{\text{out}} = 4\sigma$ (filled) and on 3–6σ (open).

first 3σ only); over a 6σ window all sizes score alike. Training on R_{out} from 3–6σ gave fewer D2 successes with lower W1 (Figure 3c); a larger region does not help D1$^+$ (0.03–0.05).

Physical checks. Successful D2 placements alone score W1 0.029 (natural 0.012, hand insertion 0.094) and survive 250 time units of MD in 87% of cases (hand insertion 91%, natural 95%); on 100 new glasses D2 success is 0.36–0.51 (hand insertion 0.23).

6 Discussion

O1–O3 are met for two of three defect types, though D2 surroundings stay about 4 times the floor. **Challenges:** runs with equal loss follow the label differently (eased by per-model w); the region disturbs the host up to its edge; placed defects are slightly less stable than natural ones; D1$^+$ needs more than a larger region. **Risks:** 3D silica is costly (565 CPU core-hours of MD for data) and its rare defects occur well under once per glass.

7 Future directions

1. **Finish generalisation** (week 10): 1,024-atom glasses and 10× slower or faster cooling.
2. **Start 3D silica** (weeks 9–13): train our conditional NequIP [1] on DM2's glasses and on 576-atom glasses made with DM2's potential [3].
3. **Fix the weaknesses**: per-model w, relaxing the region's edge, a wider label for D1$^+$.

Team contributions

Workstream	What it covers	Member
Data and physics	LAMMPS glass generation, potential and relaxation, local melt-quench baseline	Nicholaus Ladislaus
Defects and evaluation	defect rules, metrics, natural floors, bootstrap statistics, physical checks	Stacy Cheruto
Models and training	EGNN, MPNN and MLP; training runs on Colab and Kaggle	Abubakar Chilala
Sampling and conditioning	sampler, guidance, requests, w selection, region-size checks	Gentille Uwera
Baselines and writing	hand insertion, figures, reports	Michel Sangwa
Supervision	course teaching assistant; supervised the project	Boniface Godwin

GitHub. Code, datasets, trained models and every result in this report: https://github.com/chilalaabubakar/IDL-PROJECT.

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